1 Two-level configuration-conditional spectral calibration model

1.1 Purpose and scope

The goal is to learn a calibration system that is trained offline from *colocated calibration campaigns*, in which all participating sensors observe the same latent spectrum, and is then deployed online to calibrate *individual sensors* operating at arbitrary locations, where different sensors generally observe different spectra.

The model therefore has two levels:

1. an *offline identification level*, which uses same-scene multi-sensor campaigns to learn persistent sensor- and configuration-dependent calibration laws;
2. an *online deployment level*, which applies the learned persistent laws to calibrate a single sensor in an arbitrary scene, without requiring any common-field assumption at deployment time.

The key structural principle is that only the offline level uses the "all sensors see the same spectrum" assumption. The online level does not.

1.2 Offline calibration corpus

Let

$$\mathcal{D}_{\text{cal}} = \{\mathcal{C}_m\}_{m=1}^{M_{\text{cal}}}$$

be the corpus of offline calibration campaigns. Campaign m is represented by

$$\mathcal{C}_m = (\mathcal{S}_m,\, K_m,\, f_m,\, c_m,\, Y_m),$$

where:

- $\mathcal{S}_m \subseteq \{1, \dots, S_{\max}\}$ is the set of sensors participating in campaign m;
- $M_m := |\mathcal{S}_m| \geq 2$ is the number of participating sensors;
- $K_m \in \mathbb{N}$ is the number of aligned acquisitions;
- $$f_m = (f_{m,1}, \dots, f_{m,J_m}), \qquad f_{m,1} < \cdots < f_{m,J_m},$$
  is the retained common frequency grid for campaign m;

- $$c_m \in \mathcal{X}$$

  is the campaign acquisition configuration, assumed constant within the campaign;

- $$c_m = (f_{\mathrm{c},m},\, B_m,\, \mathrm{RBW}_m,\, g_{\mathrm{LNA},m},\, g_{\mathrm{VGA},m}),$$

  where $f_{\mathrm{c},m}$ is central frequency, B_m is span/sample-rate metadata, RBW_m is the resolution bandwidth, $g_{\mathrm{LNA},m}$ is the LNA gain, and $g_{\mathrm{VGA},m}$ is the VGA gain;

- $$Y_m = \{Y_{m,s,k,j}\}$$

  is the aligned PSD tensor in linear power units, indexed by

  $$s \in \mathcal{S}_m, \qquad k \in \{1, \ldots, K_m\}, \qquad j \in \{1, \ldots, J_m\}.$$

1.3 Offline same-scene observation model

For campaign m, sensor $s \in \mathcal{S}_m$, acquisition k, and frequency bin j, the observed PSD is modeled as

$$Y_{m,s,k,j} = G_{m,s,j}\, S_{m,k,j} + N_{m,s,j} + \varepsilon_{m,s,k,j}, \tag{1}$$

where:

- $S_{m,k,j} \geq 0$ is the latent common spectrum of the colocated calibration campaign at acquisition k and frequency bin j;
- $G_{m,s,j} > 0$ is the multiplicative gain of sensor s in campaign m at bin j;
- $N_{m,s,j} \geq 0$ is the additive floor of sensor s in campaign m at bin j;
- $\varepsilon_{m,s,k,j}$ is a residual term.

Conditionally on the latent spectrum and calibration parameters, the working likelihood is Gaussian:

$$\varepsilon_{m,s,k,j} \mid S_{m,k,j}, G_{m,s,j}, N_{m,s,j} \sim \mathcal{N}\Big(0, \sigma^2_{m,s,j}\Big), \tag{2}$$

with $\sigma^2_{m,s,j} > 0$.

1.4 Persistent laws and campaign-specific deviations

The two-level model separates calibration into:

1. *persistent sensor/configuration laws*, which are intended to transfer to deployment;
2. *campaign-specific deviations*, which account for residual nuisance variation during offline training.

1.4.1 Persistent latent laws

Let

$$G_s^{(0)}(f,c) > 0$$

be a nominal multiplicative baseline. In the simplest current convention one may set

$$G_s^{(0)}(f,c) \equiv 1.$$

Define three persistent latent functions:

$$\Gamma_s^\star(f,c) \in \mathbb{R}, \qquad H_s^\star(f,c) \in \mathbb{R}, \qquad V_s^\star(f,c) \in \mathbb{R},$$

where:

- $\Gamma_s^\star(f,c)$ is the persistent log-gain law;
- $H_s^\star(f,c)$ is the persistent additive-floor pre-activation;
- $V_s^\star(f,c)$ is the persistent variance pre-activation.

The persistent deployment-scale laws are then

$$G_s^\star(f,c) = G_s^{(0)}(f,c)\exp\big(\Gamma_s^\star(f,c)\big), \tag{3}$$

$$N_s^\star(f,c) = \phi\big(H_s^\star(f,c)\big), \tag{4}$$

$$\sigma_s^{2,\star}(f,c) = \sigma_{\min}^2 + \phi\big(V_s^\star(f,c)\big), \tag{5}$$

where

$$\phi(x) = \log(1+e^x)$$

is the softplus map and $\sigma_{\min}^2 > 0$ is a fixed minimum variance.

1.4.2 Neural parameterization of the persistent laws

Let $e_s \in \mathbb{R}^{d_s}$ be a trainable embedding for sensor identity s, and let

$$u_\theta : \mathcal{X} \to \mathbb{R}^{d_c}$$

be a trainable configuration encoder.

Let

$$\begin{aligned} b_\ell^\Gamma &: \mathbb{R} \to \mathbb{R}, \quad \ell = 1, \dots, L_\Gamma, \\ b_\ell^H &: \mathbb{R} \to \mathbb{R}, \quad \ell = 1, \dots, L_H, \\ b_\ell^V &: \mathbb{R} \to \mathbb{R}, \quad \ell = 1, \dots, L_V, \end{aligned}$$

be fixed smooth frequency basis functions (for example, cubic B-splines).

The persistent laws are parameterized by a neural network with global weights θ:

$$a_\theta^\Gamma(s,c) = A_\theta^\Gamma\big(e_s, u_\theta(c)\big) \in \mathbb{R}^{L_\Gamma}, \tag{6}$$

$$a_\theta^H(s,c) = A_\theta^H\big(e_s, u_\theta(c)\big) \in \mathbb{R}^{L_H}, \tag{7}$$

$$a_\theta^V(s,c) = A_\theta^V\big(e_s, u_\theta(c)\big) \in \mathbb{R}^{L_V}, \tag{8}$$

and

$$\Gamma_s^\star(f,c) = \sum_{\ell=1}^{L_\Gamma} a_{\theta,\ell}^\Gamma(s,c)\, b_\ell^\Gamma(f), \tag{9}$$

$$H_s^\star(f,c) = \sum_{\ell=1}^{L_H} a_{\theta,\ell}^H(s,c)\, b_\ell^H(f), \tag{10}$$

$$V_s^\star(f,c) = \sum_{\ell=1}^{L_V} a_{\theta,\ell}^V(s,c)\, b_\ell^V(f). \tag{11}$$

Thus one global neural system parameterized by θ generates smooth sensor- and configuration-conditional calibration laws over frequency.

1.4.3 Campaign-specific deviations

For offline campaign m, define campaign-specific residual corrections

$$\delta_{m,s,j}^\Gamma \in \mathbb{R}, \qquad \delta_{m,s,j}^H \in \mathbb{R}, \qquad \delta_{m,s,j}^V \in \mathbb{R}.$$

Then the campaign-specific parameters used in (1)–(2) are

$$G_{m,s,j} = G_s^{(0)}(f_{m,j}, c_m)\, \exp\Big(\Gamma_s^\star(f_{m,j}, c_m) + \delta^\Gamma_{m,s,j}\Big), \tag{12}$$

$$N_{m,s,j} = \phi\Big(H_s^\star(f_{m,j}, c_m) + \delta^H_{m,s,j}\Big), \tag{13}$$

$$\sigma^2_{m,s,j} = \sigma^2_{\min} + \phi\Big(V_s^\star(f_{m,j}, c_m) + \delta^V_{m,s,j}\Big). \tag{14}$$

The role of $\delta^\Gamma, \delta^H, \delta^V$ is to absorb campaign-specific nuisance variation such as local thermal conditions, residual hardware state, or other nonpersistent effects during offline identification.

1.5 Identifiability

1.5.1 Local campaign-scale nonidentifiability

For a fixed campaign m, the observation model is invariant under the transformation

$$G_{m,s,j} \mapsto a_{m,j} G_{m,s,j}, \qquad S_{m,k,j} \mapsto a_{m,j}^{-1} S_{m,k,j}, \qquad a_{m,j} > 0.$$

Hence the multiplicative scale is not identifiable without an explicit reference convention.

1.5.2 Global deployment-scale convention

To define a persistent deployment scale, choose positive weights

$$\pi_s > 0, \qquad \sum_{s=1}^{S_{\max}} \pi_s = 1,$$

and impose the global reference convention

$$\sum_{s=1}^{S_{\max}} \pi_s\, \Gamma_s^\star(f, c) = 0 \qquad \text{for all } (f, c). \tag{15}$$

Equivalently,

$$\prod_{s=1}^{S_{\max}} (G_s^\star(f, c))^{\pi_s} = \prod_{s=1}^{S_{\max}} \Big(G_s^{(0)}(f, c)\Big)^{\pi_s}.$$

If $G_s^{(0)}(f, c) \equiv 1$, then

$$\prod_{s=1}^{S_{\max}} (G_s^\star(f, c))^{\pi_s} = 1. \tag{16}$$

This fixes the persistent calibration laws on a global network reference scale. In implementation, (15) can be enforced by subtracting the weighted mean across sensors from the raw neural log-gain outputs.

1.5.3 Mean-zero campaign deviations

To prevent the campaign residual corrections from reintroducing arbitrary multiplicative scale, impose

$$\frac{1}{M_m} \sum_{s \in \mathcal{S}_m} \delta^{\Gamma}_{m,s,j} = 0, \qquad j = 1, \ldots, J_m, \quad m = 1, \ldots, M_{\text{cal}}. \tag{17}$$

Under (17), the latent common spectrum $S_{m,k,j}$ of each calibration campaign is expressed on the persistent global deployment scale defined by (15).

Remark. If an absolute metrological reference is available, then (15) may be replaced by an absolute anchor, for example by fixing

$$\Gamma^{\star}_{s_{\text{ref}}}(f, c) = 0$$

for a trusted reference instrument or by adding supervised absolute-loss terms. Without such information, the framework is rigorous only on the chosen reference scale.

1.6 Offline latent-spectrum update

For fixed values of $(\theta, \delta^{\Gamma}, \delta^{H}, \delta^{V})$, the latent common spectrum in calibration campaign m is estimated by weighted least squares.

Define

$$\omega_{m,s,j} = \frac{1}{\sigma^2_{m,s,j}}.$$

Then

$$\widehat{S}_{m,k,j} = \left[\frac{\sum_{s \in \mathcal{S}_m} \omega_{m,s,j}\, G_{m,s,j}\, (Y_{m,s,k,j} - N_{m,s,j})}{\sum_{s \in \mathcal{S}_m} \omega_{m,s,j}\, G^2_{m,s,j}} \right]_+, \tag{18}$$

where $[x]_+ = \max\{x, 0\}$ or, in implementation, clipping to a small positive numerical floor $\varepsilon_S > 0$.

1.7 Offline objective

Let

$$\Delta = \{\delta^{\Gamma}_{m,s,j}, \delta^{H}_{m,s,j}, \delta^{V}_{m,s,j}\}_{m,s,j}$$

collect all campaign-specific deviations.

1.7.1 Negative log-likelihood term

The Gaussian negative log-likelihood is

$$\mathcal{L}_{\mathrm{nll}}(\theta, \Delta, S) = \frac{1}{2} \sum_{m=1}^{M_{\mathrm{cal}}} \sum_{s \in \mathcal{S}_m} \sum_{k=1}^{K_m} \sum_{j=1}^{J_m} \left[\log \sigma_{m,s,j}^2 + \frac{\left(Y_{m,s,k,j} - G_{m,s,j} S_{m,k,j} - N_{m,s,j}\right)^2}{\sigma_{m,s,j}^2} \right]. \tag{19}$$

1.7.2 Smoothness of campaign deviations

To regularize campaign-specific corrections across frequency, define a second-difference operator $D_{m,2}$ on the frequency grid f_m. For a uniform grid it may be the usual second difference; for a nonuniform grid one should use the corresponding nonuniform second-difference operator.

The campaign-deviation regularizer is

$$\mathcal{R}_{\mathrm{dev}}(\Delta) = \sum_{m=1}^{M_{\mathrm{cal}}} \sum_{s \in \mathcal{S}_m} \left[\lambda_{\mathrm{smooth}}^{\Gamma} \| D_{m,2} \delta_{m,s,\cdot}^{\Gamma} \|_2^2 + \lambda_{\mathrm{smooth}}^{H} \| D_{m,2} \delta_{m,s,\cdot}^{H} \|_2^2 + \lambda_{\mathrm{smooth}}^{V} \| D_{m,2} \delta_{m,s,\cdot}^{V} \|_2^2 \right], \tag{20}$$

where, for example,

$$\delta_{m,s,\cdot}^{\Gamma} = (\delta_{m,s,1}^{\Gamma}, \ldots, \delta_{m,s,J_m}^{\Gamma})^{\top}.$$

1.7.3 Shrinkage of campaign deviations

Because the campaign deviations are nuisance terms, they should shrink toward zero:

$$\mathcal{R}_{\mathrm{shrink}}(\Delta) = \sum_{m=1}^{M_{\mathrm{cal}}} \sum_{s \in \mathcal{S}_m} \left[\lambda_0^{\Gamma} \| \delta_{m,s,\cdot}^{\Gamma} \|_2^2 + \lambda_0^{H} \| \delta_{m,s,\cdot}^{H} \|_2^2 + \lambda_0^{V} \| \delta_{m,s,\cdot}^{V} \|_2^2 \right]. \tag{21}$$

1.7.4 Weight decay on the persistent model

The persistent neural model is regularized by

$$\mathcal{R}_{\theta}(\theta) = \lambda_{\theta} \| \theta \|_2^2. \tag{22}$$

1.7.5 Optional reliable-sensor anchor

If campaign m provides a reliable sensor index $s_m^{\star} \in \mathcal{S}_m$, one may add a weak anchor on its campaign residual gain correction:

$$\mathcal{R}_{\mathrm{rel}}(\Delta) = \sum_{m=1}^{M_{\mathrm{cal}}} \lambda_{\mathrm{rel}} \sum_{j=1}^{J_m} \left(\delta_{m,s_m^{\star},j}^{\Gamma}\right)^2. \tag{23}$$

This is optional and should remain soft.

1.7.6 Total offline training objective

The total offline objective is

$$\mathcal{J}_{\text{off}}(\theta, \Delta, S) = \mathcal{L}_{\text{nll}}(\theta, \Delta, S) + \mathcal{R}_{\text{dev}}(\Delta) + \mathcal{R}_{\text{shrink}}(\Delta) + \mathcal{R}_{\theta}(\theta) + \mathcal{R}_{\text{rel}}(\Delta), \quad (24)$$

subject to the identifiability constraints (15) and (17).

1.8 Offline estimation problem

The offline fit is the constrained optimization problem

$$(\widehat{\theta}, \widehat{\Delta}, \widehat{S}) \in \arg\min_{\theta, \Delta, S} \mathcal{J}_{\text{off}}(\theta, \Delta, S) \quad \text{subject to (15) and (17).} \quad (25)$$

A practical implementation may use block alternation:

1. for fixed (θ, Δ), update S by (18);
2. for fixed S, update (θ, Δ) by gradient-based optimization of (24);
3. project the persistent and campaign log-gain corrections onto the constraints (15) and (17).

1.9 Persistent deployment laws after offline training

Once the offline fit is complete, the deployed persistent laws are

$$\widehat{G}_s^{\star}(f, c) = G_s^{(0)}(f, c) \exp\big(\Gamma_{s,\widehat{\theta}}^{\star}(f, c)\big), \quad (26)$$

$$\widehat{N}_s^{\star}(f, c) = \phi\big(H_{s,\widehat{\theta}}^{\star}(f, c)\big), \quad (27)$$

$$\widehat{\sigma}_s^{2,\star}(f, c) = \sigma_{\min}^2 + \phi\big(V_{s,\widehat{\theta}}^{\star}(f, c)\big). \quad (28)$$

These are the only quantities needed for online deployment in arbitrary scenes.

1.10 Online deployment on arbitrary scenes

1.10.1 Deployment data

Consider a deployed sensor $s^{\dagger} \in \{1, \ldots, S_{\max}\}$ operating under configuration $c^{\dagger}$ on a retained frequency grid

$$f^{\dagger} = (f_1^{\dagger}, \ldots, f_{J^{\dagger}}^{\dagger}).$$

Suppose it produces observations

$$Y^\dagger = \{Y^\dagger_{k,j}\}, \qquad k = 1, \ldots, K^\dagger, \quad j = 1, \ldots, J^\dagger.$$

No assumption is made that any other deployed sensor observes the same scene.

1.10.2 Deployment observation model

For deployment, the physically relevant model is

$$Y^\dagger_{k,j} = G^\dagger_j \, X^\dagger_{k,j} + N^\dagger_j + \varepsilon^\dagger_{k,j}, \tag{29}$$

where:

- $X^\dagger_{k,j}$ is the *sensor-local* latent spectrum of the deployed scene at that sensor;
- $G^\dagger_j$ and $N^\dagger_j$ are the persistent laws evaluated for that sensor and configuration;
- there is no common-spectrum relation tying this $X^\dagger$ to any other deployed sensor.

The deployed persistent calibration curves are

$$\widehat{G}^\dagger_j = \widehat{G}^\star_{s^\dagger}(f^\dagger_j, c^\dagger), \tag{30}$$

$$\widehat{N}^\dagger_j = \widehat{N}^\star_{s^\dagger}(f^\dagger_j, c^\dagger), \tag{31}$$

$$\widehat{\sigma}^{2,\dagger}_j = \widehat{\sigma}^{2,\star}_{s^\dagger}(f^\dagger_j, c^\dagger). \tag{32}$$

1.10.3 Online calibration map

The online calibrated PSD is

$$\widehat{X}^\dagger_{k,j} = \frac{[Y^\dagger_{k,j} - \widehat{N}^\dagger_j]_+}{\widehat{G}^\dagger_j}. \tag{33}$$

If one prefers unclipped subtraction, then replace $[\cdot]_+$ by identity and handle negative corrected values downstream. However, nonnegativity clipping is often numerically safer for PSD-like quantities.

1.10.4 Per-bin deployment uncertainty

A first-order variance propagation gives

$$\mathrm{Var}(\widehat{X}^{\dagger}_{k,j} \mid s^{\dagger}, c^{\dagger}) \approx \frac{\widehat{\sigma}^{2,\dagger}_{j}}{(\widehat{G}^{\dagger}_{j})^{2}}. \tag{34}$$

This is the observation-noise contribution only. It does not include epistemic uncertainty in $\widehat{\theta}$ unless such uncertainty is modeled explicitly, for example through ensembling or variational methods.

1.11 What transfers and what does not

The same-scene assumption is used only in the offline calibration corpus to identify persistent sensor/configuration behavior. It does *not* appear in the deployment rule (33).

Thus the transfer logic is:

1. offline same-scene campaigns identify

$$\widehat{G}^{\star}_{s}(f,c), \qquad \widehat{N}^{\star}_{s}(f,c), \qquad \widehat{\sigma}^{2,\star}_{s}(f,c);$$

2. online deployment applies those laws to a single sensor in an arbitrary scene, with no requirement that other sensors observe the same latent spectrum.

1.12 Optional extension: slowly varying device-state covariates

If slowly varying device-health covariates are available, such as temperature, clock error diagnostics, age, or self-test features, let

$$z_{m,s} \in \mathcal{Z}$$

denote such covariates in offline campaign m and

$$z^{\dagger} \in \mathcal{Z}$$

denote them at deployment time.

Then one may replace the persistent laws by

$$\Gamma^{\star}_{s}(f,c,z), \qquad H^{\star}_{s}(f,c,z), \qquad V^{\star}_{s}(f,c,z),$$

with the same structure as above. In that case the deployed laws become

$$\widehat{G}^{\star}_{s^\dagger}(f^\dagger_j, c^\dagger, z^\dagger), \qquad \widehat{N}^{\star}_{s^\dagger}(f^\dagger_j, c^\dagger, z^\dagger),$$

and so on. This extension is optional; if such covariates are unavailable, one simply omits z.

1.13 Summary

The two-level model consists of:

1. a global persistent neural law

$$(s, f, c) \mapsto \big(G^\star_s(f, c), N^\star_s(f, c), \sigma^{2,\star}_s(f, c)\big),$$

   learned offline from colocated same-scene campaigns;

2. campaign-specific nuisance deviations used only during offline fitting;

3. an online per-sensor calibration map

$$\widehat{X}^\dagger_{k,j} = \frac{[Y^\dagger_{k,j} - \widehat{N}^\dagger_j]_+}{\widehat{G}^\dagger_j},$$

   which remains valid even when deployed sensors observe different spectra at different locations.

Therefore the model is simultaneously:

- *configuration-conditional*;
- *sensor-specific*;
- *trained from same-scene campaigns*;
- *deployable to arbitrary scenes without a common-field assumption.*